# Programming Fundamentals Using C++

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# Week 04

## Lecture – 01

## Lecture – 02

### Variables and Data-types

**Semester Calendar**

# Class Schedule

**Lab Hours (Practice)**

**Tuesday (09:00 to 10:30)**
**Wednesday (09:30 to 12:30)**
**Wednesday (03:00 to 04:30)**

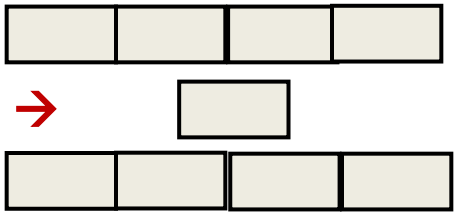
# What is a Variable?

- A variable is a memory address where data can be stored and changed.
- Declaring a variable means specifying both its name and its data type.

```
int num;
```

- What Does a Variable Declaration Do?
- A declaration tells the compiler to allocate enough memory to hold a value of this data type, and to associate the identifier with this location.

```
int ageOfDog; →
char middleInitial; →
float taxRate; →
```



# Variables:

- All variables must declared before use.
- At the top of the program
- Just before use.
- Commas are used to separate identifiers of the same type.

```
int count, age;
```

(Can declare several variables of same type in one declaration)

- Variables can be initialized to a starting value when they are declared

```
int count = 0;
int age, count = 0;
```

# What is an Identifier?

- An identifier is the name to denote labels, types, variables, constants or functions, in a C++ program.
- C++ is a case-sensitive language.

  Work is not work
- Identifiers should be descriptive
  - Using meaningful identifiers is a good programming practice
- Identifiers must be unique

# Identifiers:

- Identifiers cannot be reserved words (keywords)

  e.g. int double main return cout

- Identifier must start with a letter or underscore, and be followed by zero or more letters (A-Z, a-z), digits (0-9), or underscores

- VALID

  abc age_of_dog _taxRateY2K

  X PrintHeading ageOfHorse

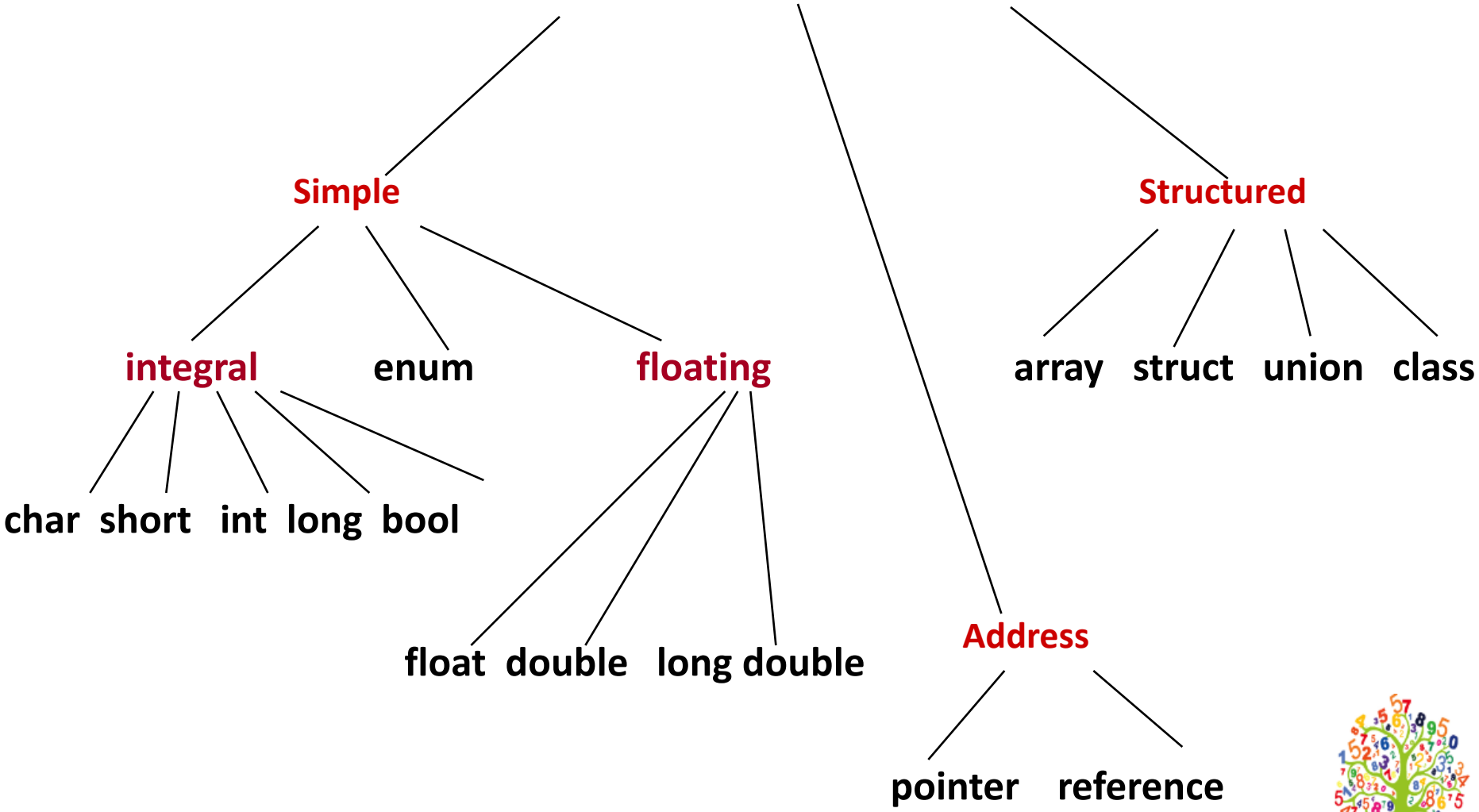
- NOT VALID

  age# 2000TaxRate Age-Of-Dog main

# Data-type

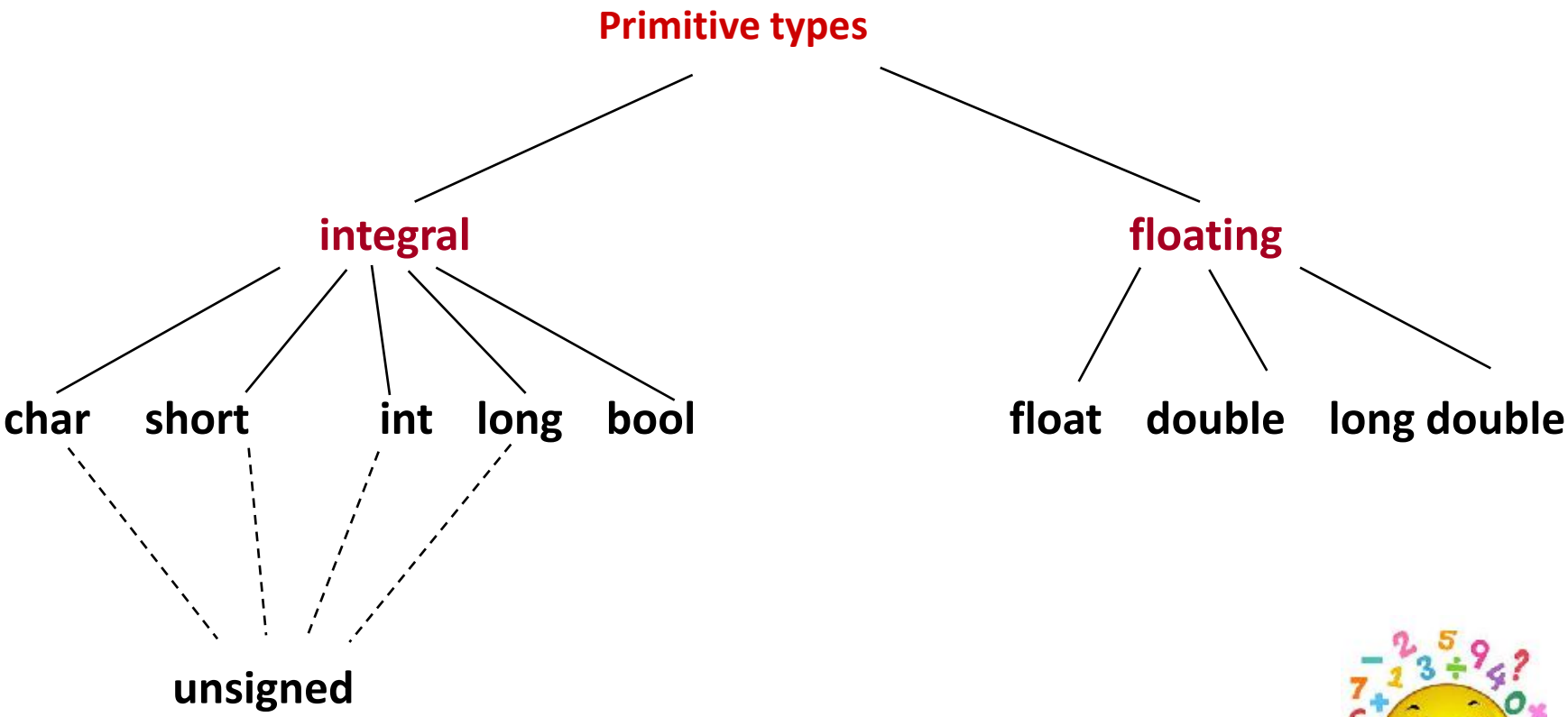
- All variables use data-type during declaration to restrict the type of data to be stored.
- Therefore, we can say that data types are used to tell the variables the type of data it can store.
- Whenever a variable is defined in C++, the compiler allocates some memory for that variable based on the data-type with which it is declared.
- Every data type requires a different amount of memory.

# C++ Data Types


Simple
Structured
integral
enum
floating
array
struct
union
class
char
short
int
long
bool
float
double
long double
Address
pointer
reference

# C++ Primitive Data Types:

Primitives data types are built-in or predefined data types and can be used directly by the user to declare variables.



# Primitive Data Types in C++

- **Integral Types** → int sample values
  **4578 -4578 0**
  - represent whole numbers and their negatives
  - declared as int, short, or long
- **Character Types** → char sample values
  **'B' 'd' '4' '?' '*'**
  - represent single characters
  - declared as char
  - Stored by ASCII values
- **Boolean Type** → bool values
  **true false**
  - declared as bool
  - has only 2 values true/false
  - will not print out directly
- **Floating Types** → float sample values
  **95.274 95.0 .265**
  - represent real numbers with a decimal point
  - declared as float, or double
  - Scientific notation where e (or E) stand for "times 10 to the " (.55-e6)

# Data Types in C++

- The data type specifies the size and type of information the variable will store

| Data Type | Size | Description |
|---|---|---|
| int | 4 bytes | Stores whole numbers, without decimals |
| float | 4 bytes | Stores fractional numbers, containing one or more decimals. Sufficient for storing 7 decimal digits |
| double | 8 bytes | Stores fractional numbers, containing one or more decimals. Sufficient for storing 15 decimal digits |
| boolean | 1 byte | Stores true or false values |
| char | 1 byte | Stores a single character/letter/number, or ASCII values |

# C++ Data Types Example

- int myNum = 5; // Integer (whole number)
- float myFloatNum = 5.99; // Floating point number
- double myDoubleNum = 9.98; // Floating point number
- char myLetter = 'D'; // Character
- bool myBoolean = true; // Boolean
- string myText = "Hello"; // String

## Size reserved by each data type

```
//cin>> Size of each data type
#include <iostream>
using namespace std;
int main() {
   cout << "Size of char : " << sizeof(char) << endl;
   cout << "Size of int : " << sizeof(int) << endl;
   cout << "Size of short int : " << sizeof(short int) << endl;
   cout << "Size of long int : " << sizeof(long int) << endl;
   cout << "Size of float : " << sizeof(float) << endl;
   cout << "Size of double : " << sizeof(double) << endl;
    return 0;
  }
```

# Integral Data types

- long b = 4523232;
- long int c = 2345342;
- long double d = 233434.56343;
- short d = 3434233; // Error! out of range
- unsigned int a = -5; // Error! can only store positive numbers or 0
- float f1 = 35e3; //A floating point number can also be a scientific
- double d1 = 12E4; //number with an "e" to indicate the power of 10:

## If Ans is Floating point

```
int main () {
    int a, b; // Variable declaration
    int c, d;
    a = 5;    // Actual initialization
    b = 2;
    c = a + b;
    cout << c << endl ;
    d = a/b;
    cout << d << endl ;
    return 0;    }
```

## If Ans is Floating point

```
int main () {
    int a, b, c; // Variable declaration
    float d;
    a = 5;    // Actual initialization
    b = 2;
    c = a + b;
    cout << c << endl ;
    d= a/b;
    cout << d << endl ;
    return 0;    }
```

# Bool Type

- A boolean data type is declared with the bool keyword and can only take the values true or false. When the value is returned, true = 1 and false = 0.

```
bool isCodingFun = true;
bool isFishTasty = false;
cout << isCodingFun;      // Outputs 1 (true)
cout << isFishTasty;      // Outputs 0 (false)
```

# Char Type

- The char data type is used to store a single character. The character must be surrounded by single quotes, like 'A' or 'c':

```
char myGrade = 'B';
cout << myGrade;
```

Guess the Output

```
{
char a = 65, b = 66, c = 67;
cout << a;
cout << b;
cout << c; }
```

# String Type

- The string type is used to store a sequence of characters (text). This is not a built-in type, but it behaves like one in its most basic usage. String values must be surrounded by double quotes

```
string greeting = "Hello";
cout << greeting;
```

Note:To use strings, you must include an additional header file in the source code, the <string> library

# Any Question